ASTRA-01 formal verification addendum

11 October 2026

The main T1–T6 obstruction statements and E1–E4 mathematical examples now have Lean 4 proof artifacts in the precise formulations summarized here. All 113 local modules passed a fresh source rebuild. This addendum updates the original research-stage statement that proof-assistant verification had not been performed. It does not establish historical novelty or an ECDLP-solving algorithm.

The 24 original research assets are preserved byte-for-byte in `original-research/`, from repository commit `31ce2acb2f90f9ca515c60fa56de01b9d576226b`. This addendum supplements the preserved manuscript; it does not replace its proofs or alter authorship.

Formal mathematical coverage

The geometric source is a genuine nonsingular Weierstrass curve, with Mathlib's actual point group and function field. The degree D is the sum of actual pole multiplicities, matching the manuscript's convention. Projective values include infinity, constructed through actual residue fields.

T1 Global obstruction
: Over algebraically closed constants in arbitrary characteristic, exact translation order $n \geq 2$, a nonconstant function and a reduced nonconstant rational update: a global projective recurrence implies update degree $r = 1$ and $n \mid D$.

T2 and T3 Finite agreement bounds
: The complete cyclic agreement set has cardinality at most $\min(n, D(r+1))$, including common poles. For a fixed finite menu, the complete union has cardinality at most $\min(n, D\sum_j (r_j + 1))$.

T4 Finite field obstruction
: For an odd prime cyclic order and a nonconstant projective recurrence, the function is injective on the cyclic subgroup; the actual PGL element has exact order n. The finite-field alternatives and minimal extension-degree divisibility are proved. The function is presented on the ambient group and restricted to that subgroup.

T5 Prime full group obstruction
: For prime $p \geq 17$, a nonsingular Weierstrass curve over $\mathbb{F}_p$ whose full point group has prime cardinality $n \neq p$, and nonzero G, every full recurrence of an arbitrary projective-valued function under a base-field PGL element forces that function constant. Hasse, exact point order, generation and representative existence are derived internally.

T6 Deterministic prime cycle
: For any deterministic update and a prime-length closed orbit, nonconstancy within the first n entries implies that those n entries are all distinct. Invertibility is not assumed.

Examples and additional identities

E1 and E2 Sharp agreement and menu examples
The explicit $\mathbb{F}_{11}$ table is identified with the actual elliptic group: 13 points, generator order 13 and $\deg(x) = 2$. There are exactly four agreements, at indices $1, 6, 8, 9$. The displayed four degree-one maps cover; every such covering menu over the base field has at least four members. Geometric base-change statements are also proved.

E3 Characteristic seven example
The actual $\mathbb{F}_7$ curve has 12 points and the specified point has order three. The degree-three y function satisfies the stated global projective semiconjugacy.

E4 Degree and evaluation cost
In characteristic different from two, the iterated doubling observable has pole degree $2{\cdot}4^t$ and value $x([2^t]P)$. Its specified recursive evaluator charges exactly t point doublings, followed by reading x.

The universal Semaev factorization and denominator-cleared addition and duplication identities are checked. Exact declarations and hypotheses are indexed in `FORMAL_VERIFICATION.md` and `lean/SEMANTIC_COVERAGE.md`; the Lean source statements take precedence over these summaries.

Trust and reproducibility

- Lean version: `4.32.2`.
- Mathlib commit: `905b95818eb32af7874a58b427f50c1711a5e96c`.
- Final fresh local-source rebuild: 113 modules, exit code 0, 535.05 seconds, peak sampled process-group RSS 3,346,960 KiB. Pinned upstream dependencies were cached; previous local objects were not reused.
- The audit records 210 printed declaration closures. The only reported axioms are `propext`, `Classical.choice`, and `Quot.sound`. No admitted proofs, custom axioms or native-evaluation proof shortcuts appear in the local source.
- Separate successful replays cover 83 modules using the same trusted compiler and dependencies. These are internal verification, not outside expert peer review.

Portable reproduction instructions, source and package checksums, successful logs, build records and axiom manifests are included. No caches or compiled objects are distributed. The audited workflow reused provisioned dependencies; clean online dependency provisioning was not exercised. Apache-2.0 attribution is retained for adapted divisor foundations.

Scope and novelty

No separate scheme-level transport theorem for every abstract curve presentation is supplied. The uniform-probability restatement of T3 follows from the cardinality bound, without a probability-space API wrapper. E4 uses a specified group-operation cost model, not a bit-complexity or general circuit lower bound. Original empirical execution histories are not certified.

The independent novelty review does not establish historical priority. Hasse and other classical prerequisites are formalized as prerequisites, not claimed as discoveries. The original U1–U5 algorithmic, complexity, sufficiency and novelty claims remain unproved or unestablished. No faster cryptographic solver or attack is supplied.